

Health & Education

Instructor Assigned Readings

The impact of artificial intelligence on learner–instructor interaction in online learning

[Kyoungwon Seo](#) [Joice Tang](#) [Ido Roll](#) [Sidney Fels](#) [Dongwook Yoon](#)

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[Adam Bohr](#) [Kaveh Memarzadeh](#)

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Paper: The impact of artificial intelligence on learner–instructor interaction in online learning**Purpose of the study**

- the **researchers** wanted to **understand**
 - how **AI** systems **change** interactions between **students** and **instructors** in online **learning**
 - how **students** and **instructors** perceive these **changes**
 - what **opportunities** and **risks** **AI** introduces in **educational** settings
- **AI systems** are increasingly used to
 - **personalize** learning
 - **automate grading** and **feedback**
 - support **communication** in online courses
- however, their effect on **human interaction in learning environments** is not yet fully understood

Key Concepts

- **instructional interaction**
 - explaining **concepts**
 - providing **feedback**
 - guiding **learning** activities
- **social interaction**
 - building **relationships**
 - encouraging **participation**
 - maintaining **engagement**
- **these interactions strongly affect**
 - student **motivation**
 - **satisfaction** with courses
 - learning **outcomes**

	AI technologies may change how these interactions occur
Research Method	- the study used a design method called “speed dating” - this method allows participants to quickly experience and evaluate different AI concepts - participants interacted with multiple hypothetical AI systems designed for online education. - examples included systems that could monitor student progress recommend learning materials automatically answer questions provide personalized feedback - researchers collected participant reactions and perceptions
Types of AI Roles in Education	- AI as a teaching assistant - AI helps instructors by grading assignments answering routine questions providing automated feedback - this reduces instructor workload - AI as a learning companion - AI directly supports students by recommending resources explaining difficult concepts guiding practice exercises - this provides personalized learning support - AI as an administrative tool - AI may handle logistical tasks such as tracking participation monitoring performance identifying struggling students - this helps instructors manage large online classes.
Key Findings	- AI can improve efficiency - participants believed AI could automate repetitive tasks

	▪ speed up feedback ▪ help instructors manage large courses • this may allow instructors to focus on higher-level teaching tasks • AI can personalize learning ○ AI systems can ▪ adapt content to individual learners ▪ provide tailored feedback ▪ recommend resources based on performance • personalization may improve student engagement and learning outcomes • human interaction is still important ○ participants emphasized that ▪ students still need human instructors ▪ emotional support and mentorship cannot easily be replaced • AI should support instructors, not replace them. • participants expressed concerns about ○ reduced human contact ▪ too much automation may reduce meaningful interaction ○ privacy and data use ▪ AI systems require large amounts of student data ○ over-reliance on technology ▪ institutions might rely too heavily on automated systems
Implications for Education	• AI should augment teaching, not replace instructors ○ best approach ○ human + AI collaboration ○ example model ▪ instructor focuses on • mentoring • complex feedback • social interaction

	▪ AI handles • routine tasks • data analysis • personalized recommendations
Future Research Needs	• artificial intelligence has the potential to transform online education • its success depends on maintaining meaningful human interaction between learners and instructors • AI works best as a support tool that enhances teaching rather than replacing it
Chapter: Artificial Intelligence in Healthcare	
Main Goal of the Chapter	• how artificial intelligence is used in education • what AI systems can do for teaching and learning • the benefits, risks, and future directions of AI-based learning systems • AI is becoming a major component of digital learning environments, especially in online and large-scale education systems
Artificial Intelligence in Education (AIED)	• AEID refers to systems that can ○ analyze student data ○ adapt learning materials ○ support instructors ○ automate educational tasks • examples include ○ intelligent tutoring systems ○ automated grading tools ○ recommendation systems for learning content ○ conversational agents or chatbots • these systems attempt to replicate or support some teaching functions

**Key Capabilities
of AI in Learning
Systems**

- **adaptive** learning
 - **analyze** student **performance**
 - **detect** knowledge **gaps**
 - **adjust** **difficulty** levels
- each student **receives** personalized **instruction**
- **intelligent tutoring**
 - **guide** students **step-by-step**
 - **give** **immediate** feedback
 - **monitor** learning **progress**
- this mimics **one-on-one tutoring**
- automated **assessment**
 - **grade** assignments
 - evaluate **short answers** or **essays**
 - **provide** **feedback** quickly
- this **reduces** instructor **workload**
- **learning analytics**
 - AI systems **analyze large** educational **datasets**
 - **identify** **struggling** students
 - **detect** learning **patterns**
 - **improve** **course design**
- these insights **help instructors** intervene **earlier**

Benefits of AI in Education	• personalization ○ customized pacing ○ adaptive difficulty ○ targeted feedback • learn according to individual needs • scalability ○ thousands of students simultaneously ○ large online courses (MOOCs) • enables mass education with individualized support • continuous feedback ○ provide feedback immediately ○ provide guidance throughout the learning process • immediate feedback improves learning efficiency • instructor support ○ grading ○ answering common questions ○ monitoring student engagement • allows instructors to focus on ○ mentoring ○ complex instruction ○ curriculum design
Limitations and Concerns	• reduced human interaction • heavy reliance on AI could ○ reduce student–teacher contact ○ weaken mentorship and emotional support • human interaction remains important for motivation and engagement • data privacy • AI systems require large amounts of student data • surveillance • misuse of personal information • data security

	• algorithmic bias • AI models may reproduce biases present in training data ○ unfair grading ○ unequal learning recommendations • over-automation ○ instructors may lose control over teaching decisions ○ students may become overly dependent on automated guidance
Future Directions for AI in Education	• human-AI collaboration ○ AI should assist instructors rather than replace them ▪ AI handles routine tasks ▪ humans handle complex teaching and mentoring • explainable AI ○ educational AI systems should be transparent ○ students and instructors should understand ▪ why recommendations are made ▪ how decisions are produced • ethical governance • AI in education requires ○ clear policies ○ ethical guidelines ○ responsible data use
Core Takeaway	• artificial intelligence can transform education by enabling ○ personalized learning ○ scalable instruction ○ automated support systems • AI should augment human teaching rather than replace • careful attention must be given to ethics, bias, and privacy

Student Leadership Group Assigned Materials

TEDx Talks. (2024, October 21). *How AI can heal healthcare* | Edmund Jackson | TEDxNashville [Video]. YouTube. <https://www.youtube.com/watch?v=7ZsyYCZB3Nw>

Lakhani, P. (2025, December 22). *This is how kids should be learning with AI* [Video]. TED Talks. https://www.ted.com/talks/priya_lakhani_this_is_how_kids_should_be_learning_with_ai

TED Talks. (2023, May 1). *How AI Could Save (Not Destroy) Education* | Sal Khan | [Video]. YouTube. <https://www.youtube.com/watch?v=hJP5GqnTrNo>

Boles, S., & Boles, S. (2025, September 19). *How AI could radically change schools by 2050*. Harvard Gazette. <https://news.harvard.edu/gazette/story/2025/09/how-ai-could-radically-change-schools-by-2050/>

Knowing the hidden costs of new tech can improve digital strategy. (2026, March 11). Healthcare IT News. <https://www.healthcareitnews.com/news/knowing-hidden-costs-new-tech-can-improve-digital-strategy>

TEDx Talk: How AI can heal healthcare

- Jackson's central argument is that healthcare is not failing because people do not care
- it is failing because modern healthcare is too complex for humans to manage consistently at scale
- he uses sepsis as a major example
 - if caught early, it can often be treated cheaply, but delays can be deadly
- he connects this to personal experience, saying sepsis killed his father, which shaped his mission
- he describes AI as something that can "wrap" healthcare systems by tracking patient status, spotting patterns, and suggesting next actions in time for clinicians to act
- he argues that AI could reduce harmful delays by reading charts, literature, and patient data in real time
- the talk also frames AI as a tool for patients and families, not just hospitals
- in his example of Martha Mills, AI could have helped her family recognize danger sooner and push for urgent care
- the broader claim is that AI should simplify healthcare complexity and help people make better decisions, not replace human care

TED Talk: *This is how kids should be learning with AI*

- **Lakhani** argues that the **current** education **model** is too one-size-fits-all.
- she says teachers are **overloaded** and often **forced** into **heavy** data and admin **work**.
- her main **warning** is that **students** can **misuse AI** as a shortcut and **mistake** smooth **output** for **real** learning
- she says **genuine** learning **requires** productive **struggle**
- she **highlights** four **learning strategies**
 - **retrieval**
 - **spacing**
 - **generation**
 - **reflection**
- her view is that **AI** should be **designed** to support those **harder forms** of learning, **not** **bypass** them
- she says **AI** can help **identify** learning **gaps**, personalize **support**, and **reduce** teacher **workload**
- but she **insists** **AI** should **expand** human **expertise**, **not** **replace** it
- her **bottom line**
 - **effort** is **not** a **flaw** in learning
 - **effort** is part of how **learning sticks**

TED Talk: *How AI Could Save (Not Destroy) Education*

- **Khan** starts with the **fear** that **students** will **use AI** to **cheat** and **stop learning**
- his **answer** is that AI can be **positive** if it is **built** with **guardrails** and **used well**
- he **argues** that every **student** could have **access** to an **AI tutor** and every **teacher** to an **AI teaching assistant**
- he **connects** this to **Bloom's** "2 sigma problem," which showed how **powerful one-to-one tutoring** can be
- he presents **AI** as a way to **scale** that kind of **support** much more **widely**
- he also **describes** **AI** as **useful** for **writing** support, **reading** support, **outlining**, **feedback**, and **deeper** subject **learning**
- for **teachers**, AI could **reduce time** spent on **lesson planning** and **routine prep work**
- his overall **position** is **optimistic**
 - **AI** does **not** have to **destroy** education
 - it could **improve learning** and **free teachers** for more **human interaction**

Harvard Gazette: *How AI could radically change schools by 2050*

- AI may **change education** so much that today's "**same lesson, same test, same pace for everyone**" model could look **outdated**
- Howard **Gardner** argues that **students** may only need a **few foundational** years in **reading, writing, math**, and some **coding**
- after that, **teachers** may **act** more like **coaches** who **guide students** toward **interests, strengths, and real-world paths**
- Anthea **Roberts** says **future learners** may need to **manage** teams of **AI tools**, more like a **director, editor, or coach** than a traditional student
- **Gardner** suggests **AI** may take over many **cognitive tasks** humans now prize, **including** some forms of disciplined **thinking, synthesis, and creativity**
- he argues that **respect** and **ethics** remain deeply **human responsibilities** and should **not** be handed to **machines**.
- the panel also **warns** that **AI** can either **expand** thinking or **replace** it, and **educators** still need to **figure out** how to **push** students **toward expansion, not cognitive laziness**

Healthcare IT News: *Knowing the hidden costs of new tech can improve digital strategy*

- they should **start** with "What **business** or **care problem** are we trying to **solve**?"
- the article **argues** that **ROI** is **local** and **context-specific**
- a **tool** that **helps** one **hospital** may **not** help **another**
- **AI** is **not** always the **best** answer
- in some places, **simpler tools** like **scheduling** or **follow-up** systems may **create** more **value**
- **leaders** should account for hidden **costs**, including **staff** time, technical **support**, **workflow** redesign, **auditing**, and **training**
- new **tools** can also **create** new **burdens**, such as **alert fatigue** and **extra data review**
- the article **recommends** longer **pilots**, often **6 to 12 months**, because **short trials** may **miss** the real long-term **impact**

- a key **theme** is that **organizations** must **measure** whether the promised **value** actually appears **after deployment**, not just **assume** it will